

Principles of Finance

Professor Robert Kosowski

MSc Climate Change, Management and Finance

Week 4



Module Overview

Week 1

Introduction, Institutions, Financial Decision making, Time value of money, Net Present Value (NPV) Rule, Cost-Benefit Analysis (**BD 1-3**)*

Week 2

Time-Value of Money (Perpetuities/Annuities), Interest Rates, Valuing Bonds (**BD 4-6**)*

Week 3

Investment Decision Rules, Fundamentals of Capital Budgeting, Valuing Stocks (**BD 7.1, 7.2 (first part), 8, 9**)* , **Saito Solar case**

Deadline for submission of group assignment 1 is 9am, Thursday, 13th November

Week 4

Capital Markets and Pricing of Risk (**BD 10**), Optimal Portfolio Choice and the CAPM (**BD 11 excl. Appendix**), Estimating the Cost of Capital (**BD 12.1-13.3, excl. Appendix**)

Week 5

Capital Structure, Debt and Taxes (**BD 12.4, 12.5, 12.6, 12.7, BD 14.1, 14.2, 14.3**), Investor Behaviour and Capital Market Efficiency (**BD 13.1, 13.2, 13.6-13.8 excl. Appendix**), Introduction to the balance sheet and asset-liability-management (ALM) of a bank, insurance company and pension fund, **M&M Pizza Case**

*(See **reading list** for detailed chapter references and what is excluded as well as additional optional readings)

Deadline for submission of group assignment 2 is 9am, Tuesday, 2nd December

Imperial means Intelligent Business

Week 3 Lectures

Learning objectives

List of topics

BD Chapter 10

- Common measures of risk and return
- Understanding Diversification, systematic return, beta and the cost of capital

BD Chapter 11

- How to calculate portfolio expected return, volatility and beta
- Differentiating Capital Market Line from Security Market Line
- Deriving the Capital Asset Pricing Model

BD Chapter 12

- How to Estimate the Market Portfolio and Beta

On a lighter note

1 week,
2 Nobel Prizes,
2 billionaires

According to Forbes
- Munger's net worth is \$2.7bn
in 2023.
- Asness' net worth is \$2.6bn in
2020

Nobel Memorial Prize in Economic Sciences /Winners (1990)



William F. Sharpe



Merton Miller



Harry Markowitz

Nobel Memorial Prize in Economic Sciences /Winners (2013)



Lars Peter Hansen



Robert J. Shiller



Eugene Fama



Clifford Asness ✓
@CliffordAsness

I have a joke about the CAPM but the securities market punchline fell flat.

1:31 am · 27 Jul 2020



"Beta and modern portfolio theory and the like – none of it makes any sense to me." Charlie Munger"

Motivating examples

We have talked about the expected cost of capital.

Now we discuss how to measure expected returns and expected risk.

Imagine sitting around a negotiating table and thinking what returns and risk people expect.

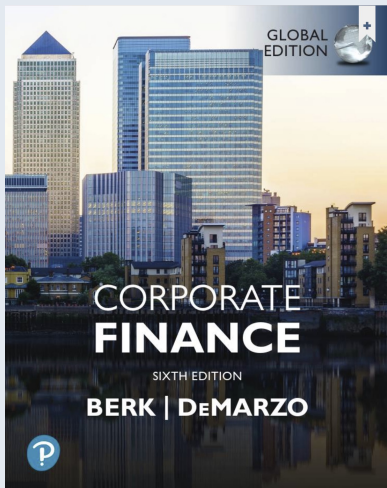
Note that these expected returns are not observable across all investors.

Note also, that these **expected** returns (“view from through the wind shield”) are conceptually not the same thing as past historical returns (“rearview mirror”) or future realized returns (“what we later see around the corner”).

So this makes it difficult to measure expected returns and relate them to expected risk measures.



Chapter 10 (BD)



Chapter Outline

- 10.2** Common Measures of Risk and Return
- 10.3** Historical Returns of Stocks and Bonds
- 10.4** The Historical Tradeoff Between Risk and Return
- 10.5** Common Versus Independent Risk
- 10.6** Diversification in Stock Portfolios
- 10.7** Measuring Systematic Risk
- 10.8** Beta and the Cost of Capital

Chapter 10

Learning Objectives

- Define a probability distribution, the mean, the variance, the standard deviation, and the volatility.
- Define and contrast idiosyncratic and systematic risk and the risk premium required for taking each on.
- Define an efficient portfolio and a market portfolio.
- Discuss how beta can be used to measure the systematic risk of a security.
- Use the Capital Asset Pricing Model to calculate the expected return for a risky security.
- Use the Capital Asset Pricing Model to calculate the cost of capital for a particular project.
- Explain why in an efficient capital market the cost of capital depends on systematic risk rather than diversifiable risk.

10.2 Common Measures of Risk and Return

Table 10.1 Probability Distribution of Returns for BFI

- Probability Distributions
 - When an investment is risky, it may earn different returns
 - Each possible return has some likelihood of occurring
 - This information is summarized with a probability distribution, which assigns a probability, P_R , that each possible return, R , will occur
 - Assume BFI stock currently trades for \$100 per share
 - In one year, there is a 25% chance the share price will be \$140, a 50% chance it will be \$110, and a 25% chance it will be \$80

Current Stock Price (\$)	Stock Price in One Year (\$)	Probability Distribution	
		Return, R	Probability, p_R
100	140	0.40	25%
	110	0.10	50%
	80	-0.20	25%

- Expected (Mean) Return
 - Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities. Negative Weights correspond to **Short Sales**.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

$$E[R_{\text{BFI}}] =$$

Variance and Standard Deviation

Variance

The expected squared deviation from the mean

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

Standard Deviation

The square root of the variance

$$\text{SD}(R) = \sqrt{\text{Var}(R)}$$

- Both are measures of the risk of a probability distribution

For BFI, the variance and standard deviation are

$$\text{Var}[R_{\text{BFI}}] =$$

$$\text{SD}(R) =$$

In finance, the standard deviation of a return is also referred to as its **volatility**

The standard deviation is easier to interpret because it is in the same units as the returns themselves

Historical Returns of Stocks and Bonds

By counting the number of times a realized return falls within a particular range, we can estimate the underlying (empirical) probability distribution

Table 10.2 Realized Return for the S&P 500, Microsoft, and Treasury Bills, 2005– 2021

Year End	S&P 500 Index	Dividends Paid*	S&P 500 Realized Return	Microsoft Realized Return	1-Month T-Bill Return
2004	1211.92				
2005	1248.29	23.15	4.9%	−0.9%	3.0%
2006	1418.30	27.16	15.8%	15.8%	4.8%
2007	1468.36	27.86	5.5%	20.8%	4.7%
2008	903.25	21.85	−37.0%	−44.4%	1.5%
2009	1115.10	27.18	26.5%	60.5%	0.1%
2010	1257.64	25.44	15.1%	−6.5%	0.1%
2011	1257.60	26.60	2.1%	−4.5%	0.0%
2012	1426.19	32.67	16.0%	5.8%	0.1%
2013	1848.36	39.75	32.4%	44.3%	0.0%
2014	2058.90	42.47	13.7%	27.6%	0.0%
2015	2043.94	43.45	1.4%	22.7%	0.0%
2016	2238.83	49.56	12.0%	15.1%	0.2%
2017	2673.61	53.99	21.8%	40.7%	0.8%
2018	2506.85	49.54	−4.4%	20.8%	1.7%
2019	3230.78	65.39	31.5%	57.6%	2.1%
2020	3756.07	69.14	18.4%	42.5%	0.4%
2021	4766.18	68.08	28.7%	52.5%	0.0%

$$\text{Average Annual Return } \bar{R} = \frac{1}{T} (R_1 + R_2 + \dots + R_T) = \frac{1}{T} \sum_{t=1}^T R_t$$

Variance Estimate Using Realized Returns

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Determining Covariance and Correlation

- Covariance

- The expected product of the deviations of two returns from their means.
- Covariance between Returns R_i and R_j

$$\text{Cov}(R_i, R_j) = E[(R_i - E[R_i]) (R_j - E[R_j])]$$

- Estimate of the Covariance from Historical Data

$$\text{Cov}(R_i, R_j) = \frac{1}{T - 1} \sum_t (R_{i,t} - \bar{R}_i) (R_{j,t} - \bar{R}_j)$$

- If the covariance is positive, the two returns tend to move together.
- If the covariance is negative, the two returns tend to move in opposite directions.

- Correlation

- A measure of the common risk shared by stocks that does not depend on their volatility

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\text{SD}(R_i) \text{SD}(R_j)}$$

- The correlation between two stocks will always be between -1 and $+1$.

Concept check

Problem

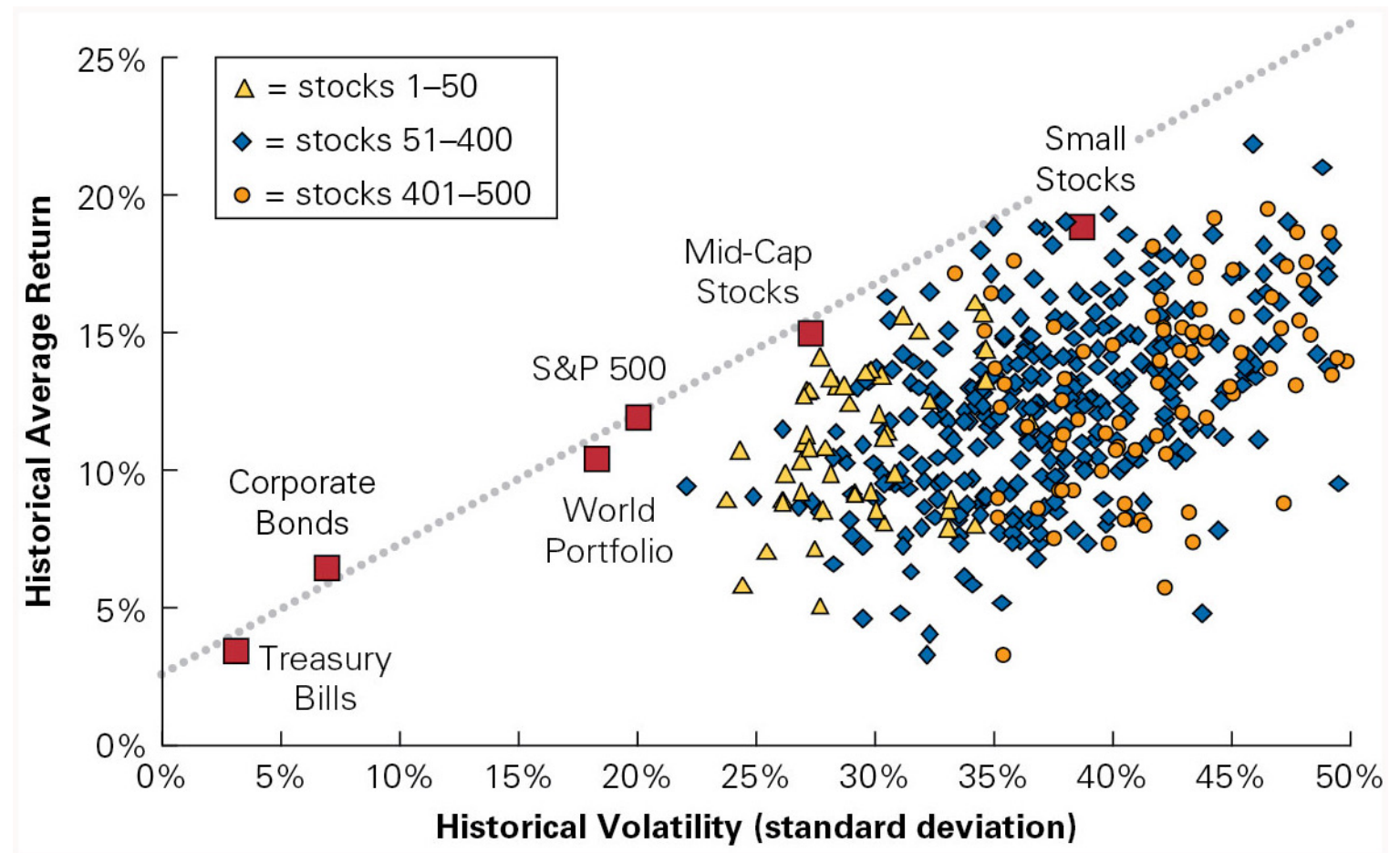
Calculate the **mean**, **variance**, **standard deviation**, **covariance** and **correlation** for the following return series for two assets 1 and 2 which each have three observations:

- $r_1 = 4, 5, 6$ and
- $r_2 = 6, 7, 8$

What is the correlation coefficient between the two assets?

Solution

Figure 10.7 Historical Volatility and Return for 500 Individual Stocks, Ranked Annually by Size



Market efficiency summary

Chapter 9.5 Information,
Competition and Stock Prices

The idea that markets aggregate the information of many investors, and that this information is reflected in security prices, is a natural consequence of investor competition.

The idea that competition among investors works to eliminate all positive-NPV trading opportunities is referred to as the **efficient markets hypothesis** (EMH) that Nobel laureate Eugene Fama is known for among other contributions.

It implies that

- securities will be fairly priced, based on their future cash flows, given all information that is available to investors.
- there is a relationship between returns – themselves functions of prices as we know – and risk – which is one form of information.

To test the EMH, we can distinguish different forms of information

1. Public Information (FT, WSJ, Bloomberg, ...)
2. Private Information (Insiders such as Apple CEO **Tim Cook**,...)



10.5 Common Versus Independent Risk

10.6 Diversification in Stock Portfolios



- Common Risk
 - Risk that is perfectly correlated
 - Risk that affects all securities
- Independent Risk
 - Risk that is uncorrelated
 - Risk that affects a particular security
- Diversification
 - The averaging out of independent risks in a large portfolio
- (A) Firm-Specific Versus (B) Systematic Risk
 - Firm Specific News
 - Good or bad news about an individual company
 - Market-Wide News
 - News that affects all stocks, such as news about the economy
 - **Synonyms:**
 - (A) Firm-Specific Risk, Idiosyncratic Risk, Unique Risk, Unsystematic Risk, Diversifiable Risk
 - (B) Systematic Risk, Undiversifiable Risk, Market Risk

No Arbitrage and the Risk Premium



10.6 Diversification in Stock Portfolios

- **The risk premium of a security is determined by its systematic risk and does not depend on its diversifiable risk**
 - This implies that a stock's volatility, which is a measure of total risk (that is, systematic risk plus diversifiable risk), is not especially useful in determining the risk premium that investors will earn
 - Standard deviation is not an appropriate measure of risk for an individual security
 - There should be no clear relationship between volatility and average returns for individual securities
 - Consequently, to estimate a security's expected return, we need to find a measure of a security's systematic risk

Measuring Systematic Risk

- Efficient Portfolio
 - A portfolio that contains only systematic risk
 - There is no way to reduce the volatility of the portfolio without lowering its expected return
- Market Portfolio
 - An efficient portfolio that contains all shares and securities in the market
 - The S&P 500 is often used as a proxy for the market portfolio
- Sensitivity to Systematic Risk: Beta (β)
 - **The expected percent change in the excess return of a security for a 1% change in the excess return of the market portfolio**
 - Beta differs from volatility. Volatility measures total risk (systematic plus unsystematic risk), while beta is a measure of only systematic risk

Problem

Suppose the market portfolio tends to increase by 47% when the economy is strong and decline by 25% when the economy is weak. Assume a 50-50 chance of the economy being either strong or weak. **What is the beta of a type S firm** whose return is 40% on average when the economy is strong and -20% when the economy is weak? What is the beta of a type I firm that bears only idiosyncratic, firm-specific risk?

Measuring Systematic Risk

- **Solution**

The systematic risk of the strength of the economy produces a.....

change in the return of the market portfolio.

The type S firm's return changes by

on average. Thus the firm's beta is

That is, each..... change in the return of the market portfolio leads to achange in the type S firm's return on average.

Beta and the Cost of Capital

- Estimating the Risk Premium
 - Market risk premium
 - The market risk premium is the reward investors expect to earn for holding a portfolio with a beta of 1

$$\text{Market Risk Premium} = E[R_{\text{Mkt}}] - r_f$$

- Adjusting for Beta
 - Estimating a Traded Security's Cost of Capital of an investment from Its Beta

$$\begin{aligned} E[R] &= \text{Risk-Free Interest Rate} + \text{Risk Premium} \\ &= r_f + \beta \times (E[R_{\text{Mkt}}] - r_f) \end{aligned} \quad (\text{BD Eq. 10.11})$$

- Equation 10.11 is often referred to as the **Capital Asset Pricing Model (CAPM)**
- It is the most important method for estimating the cost of capital that is used in practice

Interpreting Beta

Table 10.6. Betas with Respect to the S&P 500 for Individual Stocks (based on weekly data for 2017– 2022)

Company	Ticker	Industry	Equity Beta
Hormel Foods	HRL	Packaged Foods	0.16
Clorox	CLX	Household Products	0.18
SPDR Gold	GLD	Gold ETF	0.18
Campbell Soup	CPB	Packaged Foods	0.27
Johnson & Johnson	JNJ	Pharmaceuticals	0.60
Pfizer	PFE	Pharmaceuticals	0.67
Amgen	AMGN	Biotechnology	0.72
Pepsico	PEP	Soft Drinks	0.73
McDonald's	MCD	Restaurants	0.78
NVIDIA	NVDA	Semiconductors	1.53
Brunswick	BC	Leisure Products	1.60
Macy's	M	Department Stores	1.61
Nordstrom	JWN	Department Stores	1.69
Tesla	TSLA	Automobile Manufacturers	1.70
Alcoa	AA	Aluminum	1.89
Wynn Resorts Ltd.	WYNN	Casinos and Gaming	1.92
United Airlines	UAL	Airlines	1.97
Avis Budget Group	CAR	Car Rental	2.55

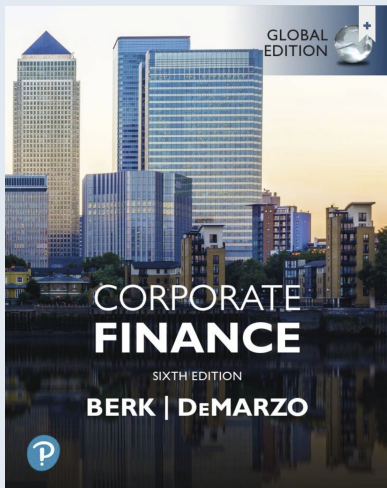
Example

- Assume
 - Advanced Micro Devices (AMD) has a beta of 1.51
 - Procter and Gamble (PG) has a beta of 0.56
 - risk-free interest rate is 4%
 - Market risk premium is 5%
- Calculate the equity cost of capital for both firms using the CAPM
- The equity cost of capital for
 - AMD is:
 - PG is:

Chapter 10 Quiz

1. How do we calculate the expected return of a stock?
2. What are the two most common measures of risk, and how are they related to each other?
3. Do expected returns for individual stocks appear to increase with volatility?
4. What is the difference between common risk and independent risk?
5. Explain why the risk premium of diversifiable risk is zero.
6. Why is the risk premium of a security determined only by its systematic risk?
7. Define the beta of a security.
8. How can you use a security's beta to estimate its cost of capital?
9. If a risky investment has a beta of zero, what should its cost of capital be according to the CAPM? How can you justify this?

Chapter 11 (BD)



Chapter Outline

- 11.1** The Expected Return of a Portfolio
- 11.2** The Volatility of a Two-Stock Portfolio
- 11.3** The Volatility of a Large Portfolio
- 11.4** Risk Versus Return: Choosing an Efficient Portfolio
- 11.5** Risk-Free Saving and Borrowing
- 11.6** The Efficient Portfolio and Required Returns
- 11.7** The Capital Asset Pricing Model
- 11.8** Determining the Risk Premium

Chapter 11

Learning Objectives

- Given a portfolio of stocks, including the holdings in each stock and the expected return in each stock, compute the following:
 - **Expected return** on the portfolio (Eq. 11.3)
 - **Covariance** of each pair of stocks in the portfolio (Eq. 11.5)
 - **Correlation coefficient** of each pair of stocks in the portfolio (Eq. 11.6)
- **Variance** of the portfolio (Eq. 11.8) and standard deviation of the portfolio
- Illustrate why the risk-return combinations of the risk-free investment and a risky portfolio lie on a straight line.
- Define the **Sharpe ratio**, and explain how it helps identify the portfolio with the highest possible expected return for any level of volatility, and how this information can be used to identify the tangency (efficient) portfolio
- Calculate the **beta** of investment with a portfolio.
- Use the beta of a security, expected return on a portfolio, and the risk-free rate to decide whether buying shares of that security will improve the performance of the portfolio.
- Compare and contrast the **capital market line** with the **security market line**.
- Define beta for an individual stock and for a portfolio.

The Expected Return of a Portfolio

- Then the return on the portfolio, R_p , is the weighted average of the returns on the investments in the portfolio, where the weights correspond to portfolio weights.

$$R_p = X_1 R_1 + X_2 R_2 + \cdots + X_n R_n = \sum_i X_i R_i$$

- The expected return of a portfolio is the weighted average of the expected returns of the investments within it.

$$E[R_p] = E\left[\sum_i X_i R_i\right] = \sum_i E[X_i R_i] = \sum_i X_i E[R_i]$$

- The variance of a portfolio is equal to the weighted average covariance of each stock with the portfolio:

$$\text{Var}(R_p) = \text{Cov}(R_p, R_p) = \text{Cov}\left(\sum_i X_i R_i, R_p\right) = \sum_i X_i \text{Cov}(R_i, R_p)$$

– which reduces to

$$\begin{aligned}\text{Var}(R_p) &= \sum_i X_i \text{Cov}(R_i, R_p) = \sum_i X_i \text{Cov}(R_i, \sum_j X_j R_j) \quad (BD \text{ Equation } 11.11) \\ &= \sum_i \sum_j X_i X_j \text{Cov}(R_i, R_j)\end{aligned}$$

The Volatility of a Portfolio

Concept Check

Problem

Calculate the **portfolio** mean and standard deviation for an **equal-weighted** portfolio consisting of two assets (1 and 2) which each have three observations for the **same** three month period:

- $r_1 = 4, 5, 6$ and
- $r_2 = 6, 7, 8$

Solution

Diversification with an Equally Weighted Portfolio

- Equally Weighted Portfolio
 - A portfolio in which the same amount is invested in each stock
- Variance of an Equally Weighted Portfolio of n Stocks

$$\text{Var}(R_P) = \frac{1}{n} (\text{Average Variance of the Individual Stocks}) + \left(1 - \frac{1}{n}\right) (\text{Average Covariance between the Stocks}) \quad (\text{BD Eq. 11.12})$$

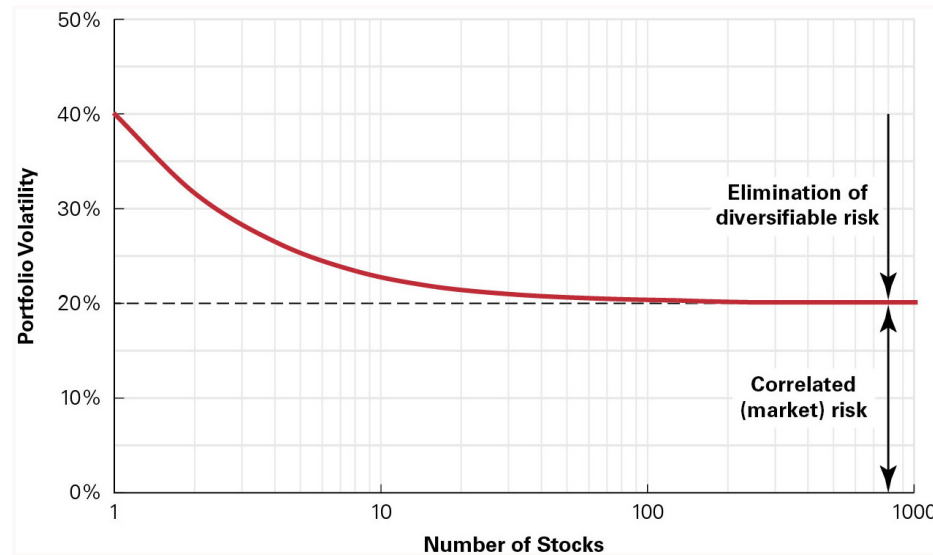


Figure 11.2 Volatility of an Equally Weighted Portfolio Versus the Number of Stocks

To derive equation 11.12 apply Equation 11.11 to an equally weighted portfolio and check how many variance and covariance terms there are.

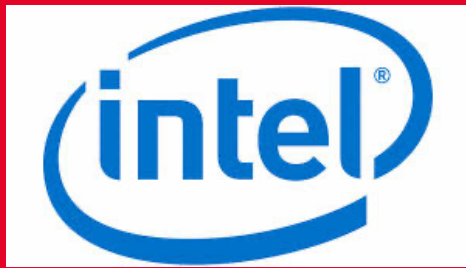
11.4 Risk Versus Return: Choosing an Efficient Portfolio

- Efficient Portfolios with Two Stocks
 - Identifying Inefficient Portfolios
 - In an **inefficient portfolio**, it is possible to find another portfolio that is better in terms of both expected return and volatility.
 - Identifying Efficient Portfolios
 - Recall from Chapter 10, in an **efficient portfolio** there is no way to reduce the volatility of the portfolio without lowering its expected return.
- Efficient Portfolios with Two Stocks
 - Consider a portfolio of Intel and Coca-Cola.

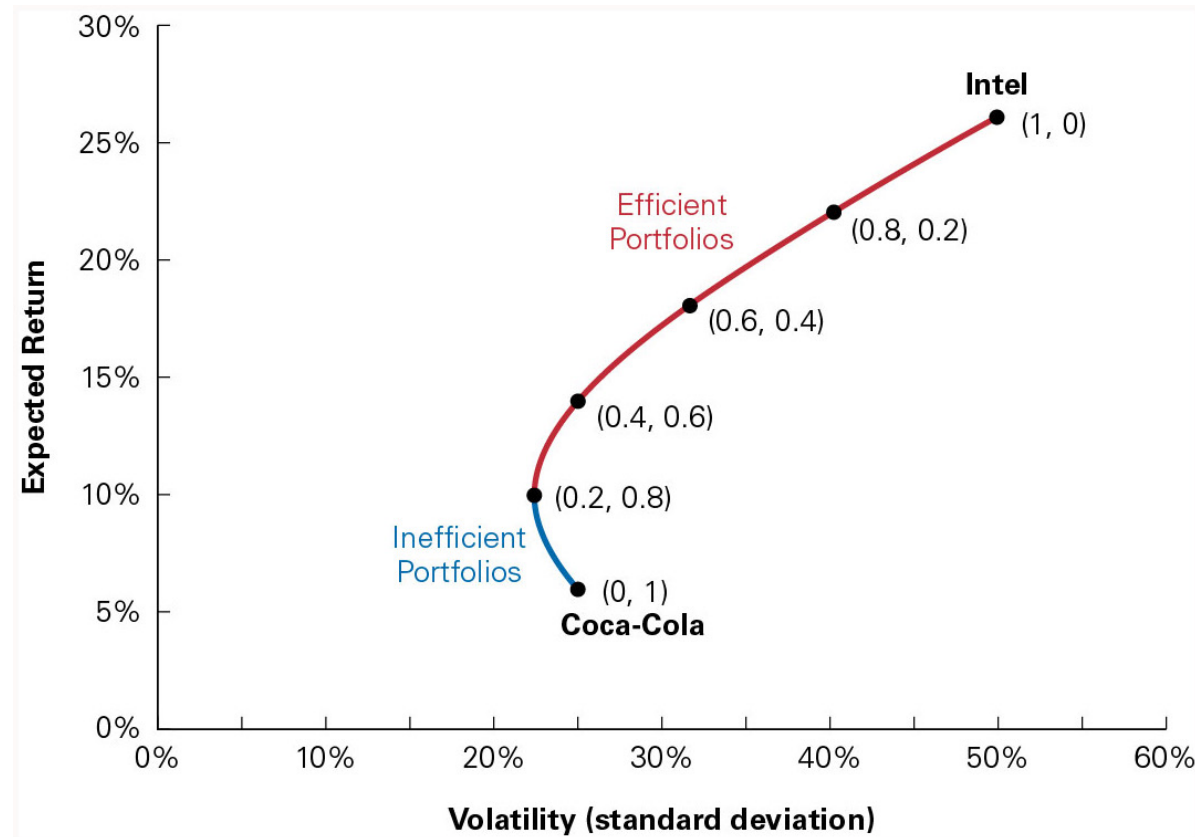
Table 11.4 Expected Returns and Volatility for Different Portfolios of Two Stocks

Portfolio Weights		Expected Return (%)	Volatility (%)
x_I	x_C	$E[R_p]$	$SD[R_p]$
1.00	0.00	26.0	50.0
0.80	0.20	22.0	40.3
0.60	0.40	18.0	31.6
0.40	0.60	14.0	25.0
0.20	0.80	10.0	22.4
0.00	1.00	6.0	25.0

Figure 11.3 Volatility Versus Expected Return for Portfolios of Intel and Coca-Cola Stock



Textbook Example 11.9



Problem

Sally Ferson has invested 100% of her money in coca-cola stock and is seeking investment advice. She would like to earn the highest expected return possible without increasing her volatility. Which portfolio would you recommend?

Textbook Example 11.9

Solution

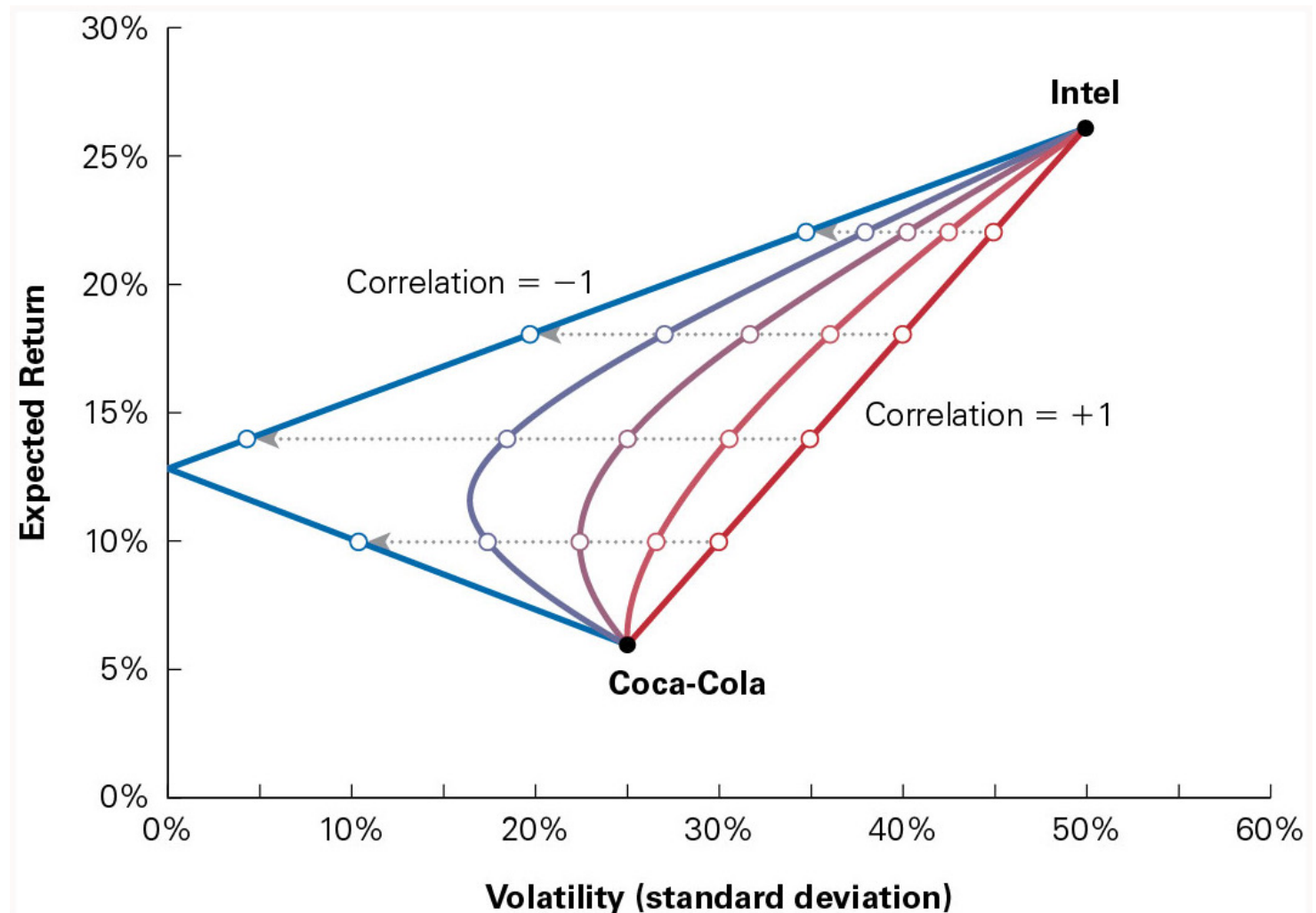
In Figure 11.3, we can see that Sally can invest up to in Intel stock without increasing her volatility.

Because stock has a higher expected return than coca-cola stock, she will earn higher expected returns by putting more money in Intel stock.

Therefore, you should recommend that sally putof her money in Intel stock, leaving in coca-cola stock.

This portfolio has the same volatility of but an expected return ofrather than theshe has now.

Figure 11.4 Effect on Volatility and Expected Return of Changing the Correlation Between Intel and Coca-Cola Stock

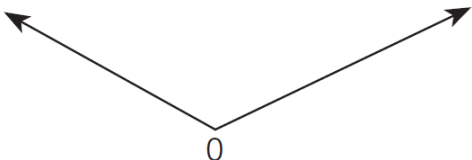


Investing in Risk-Free Securities

- Consider an arbitrary risky portfolio and the effect on risk and return of putting a fraction of the money in the portfolio, while leaving the remaining fraction in risk-free Treasury bills.
 - The expected return would be

$$\begin{aligned} E[R_{xP}] &= (1 - x)r_f + xE[R_P] \\ &= r_f + x(E[R_P] - r_f) \end{aligned}$$

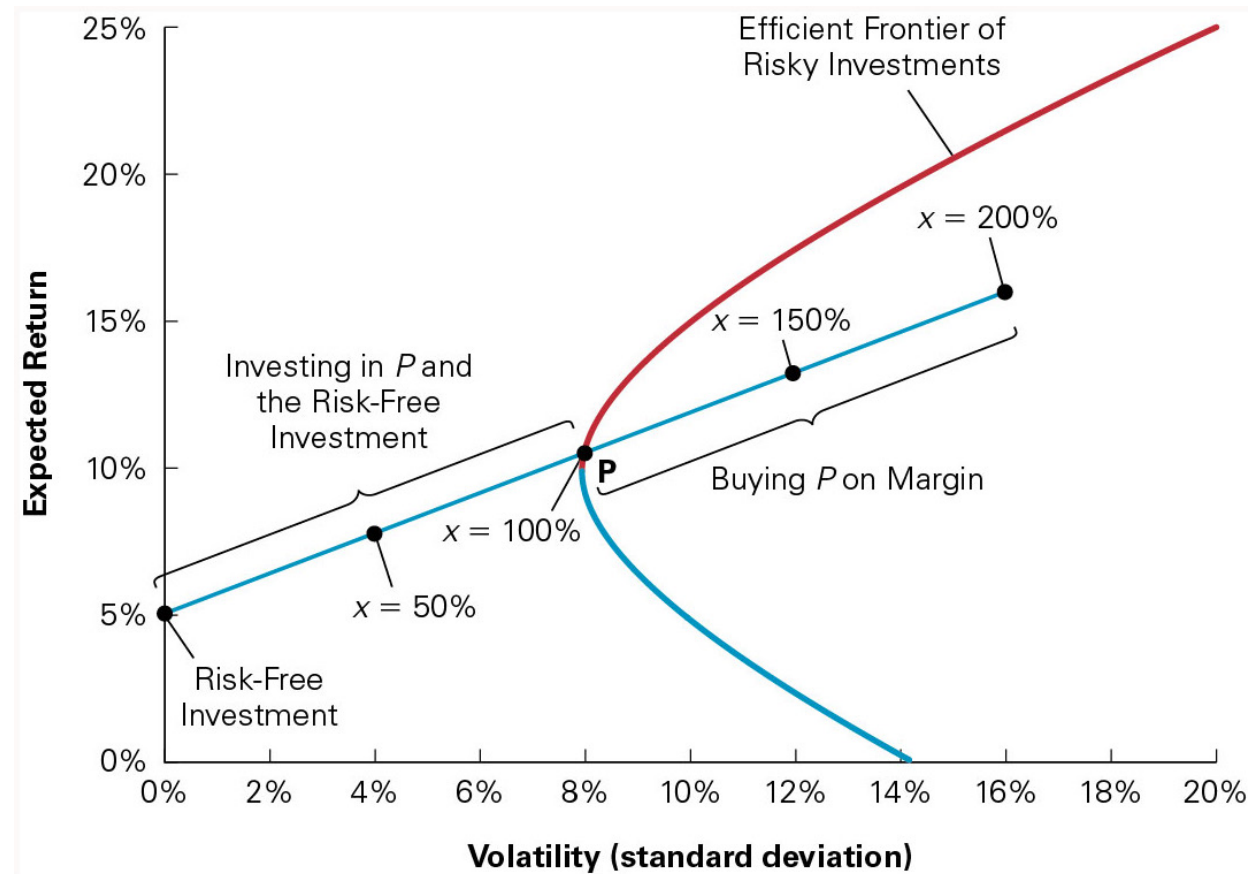
- The standard deviation of the portfolio would be calculated as follows:

$$\begin{aligned} SD(R_{xP}) &= \sqrt{(1 - x)^2 Var(r_f) + x^2 Var(R_P) + 2(1 - x)x Cov(r_f, R_P)} \\ &= \sqrt{x^2 Var(R_P)} \\ &= x SD(R_P) \end{aligned} \tag{11.16}$$


- The above implies that a portfolio that contains both the risk free asset and a risky portfolio has an expected return and standard deviation that are linear in the weight of the risky portion.

Figure 11.9 The Risk–Return Combinations from Combining a Risk-Free Investment and a Risky Portfolio

Buying on margin is equivalent to borrowing



Sharpe Ratio

Measures the ratio of reward-to-volatility provided by a portfolio

$$\text{Sharpe Ratio} = \frac{\text{Portfolio Excess Return}}{\text{Portfolio Volatility}} = \frac{E[R_P] - r_f}{SD(R_P)}$$

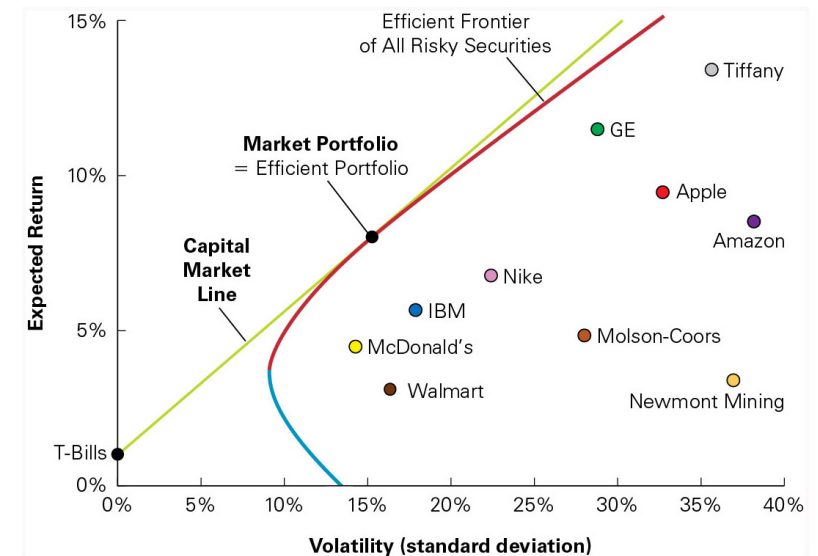
The Capital Market Line

- The expected return and volatility of a capital market line portfolio are

$$E[R_{xCML}] = (1 - x)r_f + xE[R_{Mkt}] = r_f + x(E[R_{Mkt}] - r_f)$$

$$SD(R_{xCML}) = xSD(R_{Mkt})$$

- The portfolio with the highest Sharpe ratio is the portfolio where the line with the risk-free investment is tangent to the efficient frontier of risky investments.
- The portfolio that generates this tangent line is known as the **tangent portfolio**.



- This means that the tangent portfolio is efficient and that all efficient portfolios are combinations of the risk-free investment and the tangent portfolio.
- Every investor should invest in the tangent portfolio **independent of his or her taste for risk**.

11.7 The Capital Asset Pricing Model

Three Main Assumptions

- Three Main Assumptions
 - Assumption 1
 - Investors can buy and sell all securities at competitive market prices (without incurring taxes or transaction costs) and can borrow and lend at the risk-free interest rate.
 - Assumption 2
 - Investors hold only efficient portfolios of traded securities—portfolios that yield the maximum expected return for a given level of volatility.
 - Assumption 3
 - Investors have **homogeneous expectations** regarding the volatilities, correlations, and expected returns of securities.
 - Homogeneous Expectations
 - All investors have the same estimates concerning future investments and returns.
- If investors have homogeneous expectations, then each investor will identify the same portfolio as having the highest Sharpe ratio in the economy.
- Thus, all investors will demand the same efficient portfolio of risky securities— the tangent portfolio in Figure 11.11—

- CML:

- The CML depicts portfolios combining the risk-free investment and the efficient portfolio, and shows the highest expected return that we can attain for each level of volatility. According to the CAPM, the market portfolio is on the CML and all other stocks and portfolios contain diversifiable risk and lie to the right

- SML

- According to the CAPM there is a linear relationship between a stock's beta and its expected return
- The **security market line (SML)** is graphed as the line through the risk-free investment and the market.

Figure 11.11 The Capital Market Line(CML)

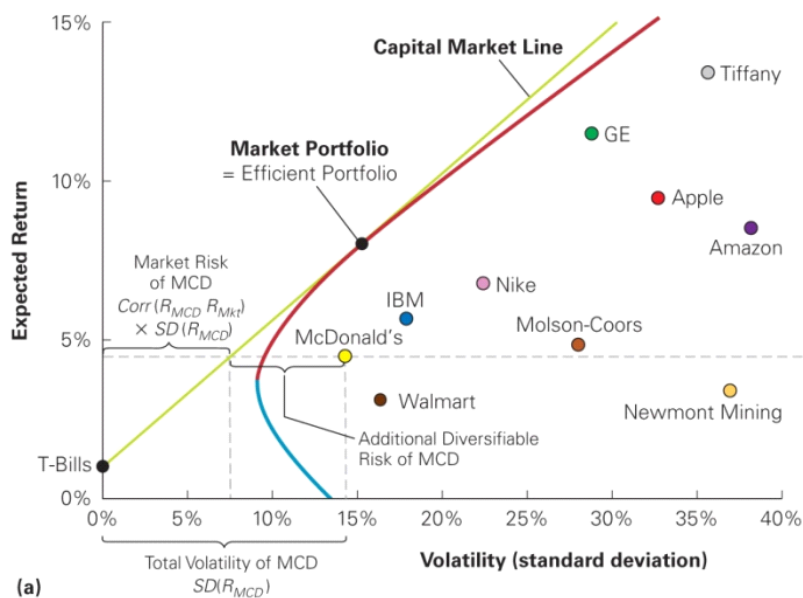
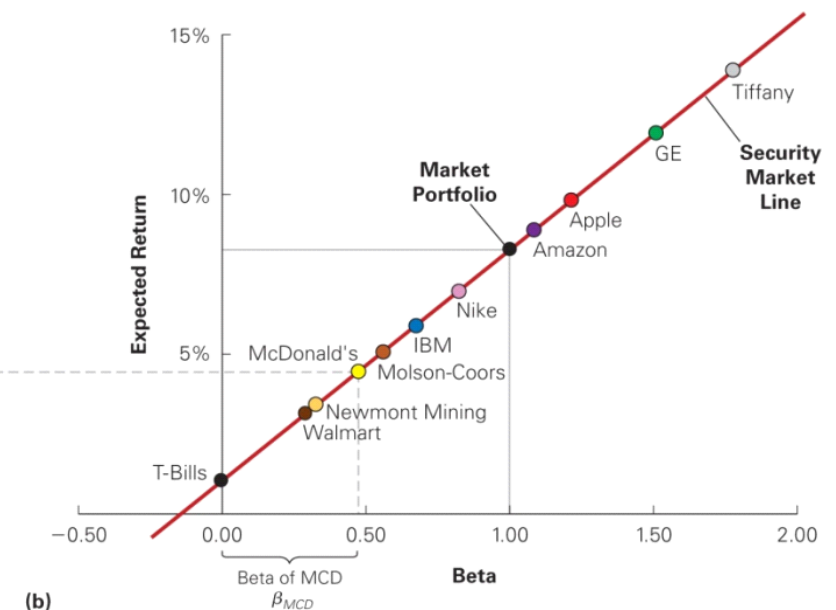


Figure 11.12 the Security Market Line (SML)



11.8 Determining the Risk Premium

- Market Risk and Beta

- Given an efficient market portfolio, the expected return of an investment is

$$E[R_i] = r_i = r_f + \beta_i(E[R_{Mkt}] - r_f)$$

Risk premium for security i

- Note this is a statement about the **cross-section**

- The beta is defined as follows:

$$\beta_i = \frac{\overbrace{SD(R_i) \times Corr(R_i, R_{Mkt})}^{\text{Volatility of } i \text{ that is common with the market}}}{SD(R_{Mkt})} = \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})}$$

- Note that beta is calculated using **time-series** data

Computing the Expected Return for a stock

Textbook Example 11.16



Problem

Suppose the risk-free return is 4% and the market portfolio has an expected return of 10% and a volatility of 16%.

3M stock has a 22% volatility and a correlation with the market of 0.50.

What is 3M's beta with the market?

What capital market line portfolio has equivalent market risk, and what is the expected return?

What is the Sharpe Ratio of 3M stock if you use the expected return you calculated?

Computing the Expected Return for a stock

Textbook Example 11.16

Solution

- We can compute beta using Eq. 11.23:

$$\beta_{MMM} =$$

- That is, for eachmove of the market portfolio, 3M stock tends to move

- We could obtain the same sensitivity to market risk by investing in the market portfolio, and in the risk-free security.

- Because it has the same market risk, 3M's stock should have the same expected return as this portfolio, which is (using Eq. 11.15 with $x = \dots$),

$$E[R_{MMM}] = \dots$$

- Because $x = \beta_{MMM}$, this calculation is precisely the CAPM Eq. 11.22. Thus, investors will require an expected return of to compensate for the risk associated with 3M stock.

- Sharpe Ratio =

Beta of a Portfolio

- Beta of a Portfolio
 - The beta of a portfolio is the weighted average beta of the securities in the portfolio.

$$\beta_P = \frac{\text{Cov}(R_P, R_{Mkt})}{\text{Var}(R_{Mkt})} = \frac{\text{Cov}(\sum_i X_i R_i, R_{Mkt})}{\text{Var}(R_{Mkt})} = \sum_i X_i \frac{\text{Cov}(R_i, R_{Mkt})}{\text{Var}(R_{Mkt})} = \sum_i X_i \beta_i \quad (\text{BD Eq. 1.24})$$

Example 11.18

Textbook Example 11.18

Problem

Suppose **Kraft Foods**' stock has a beta of 0.50, whereas **Boeing**'s beta is 1.25.

If the risk-free rate is 4%, and the expected return of the market portfolio is 10%, what is the expected return of an equally weighted portfolio of Kraft Foods and Boeing stocks, according to the CAPM?

Beta of a Portfolio

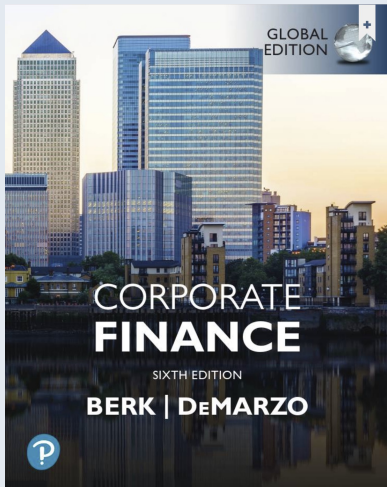
Solution

Example 11.18

Chapter 11 Quiz

1. How do you calculate a portfolio return?
2. Conceptually, how are covariance and correlation different?
3. How does adding stocks to a portfolio affect its volatility?
4. What is the efficient frontier?
5. What is the Sharpe ratio, and what does it measure?
6. How does beta affect an investment's cost of capital?
7. How are the capital market line and security market line similar? How are they different?

Chapter 12 (BD)



Chapter Outline

12.1 The Equity Cost of Capital

12.2 The Market Portfolio

12.3 Beta Estimation

Chapter 12.1-12.3 Learning Objectives

- Estimate a company's cost of capital using the CAPM equation for the Security Market Line.
- Describe the market portfolio and how it is constructed in practice.
- Discuss the attributes of a value-weighted portfolio.
- Describe common proxies for the market return and the risk-free rate.
- Define alpha and beta and explain how they are generally estimated.
- Compare the use of average return versus beta and the SML to estimate cost of equity capital.

12.2 The Market Risk Premium

To apply the CAPM, we must identify the market portfolio and risk free rate

Table 12.1 Historical Excess Returns of the S&P 500 Compared to One-Year and Ten-Year U.S. Treasury Securities

Constructing the Market Portfolio

- Market Capitalization
 - The total market value of a firm's outstanding shares
- $$MV_i = (\text{Number of Shares of } i \text{ Outstanding}) \times (\text{Price of } i \text{ per Share}) = N_i \times P_i$$
- Value-Weighted Portfolio
 - A portfolio in which each security is held in proportion to its market capitalization

$$X_i = \frac{\text{Market Value of } i}{\text{Total Market Value of All Securities}} = \frac{MV_i}{\sum_j MV_j}$$

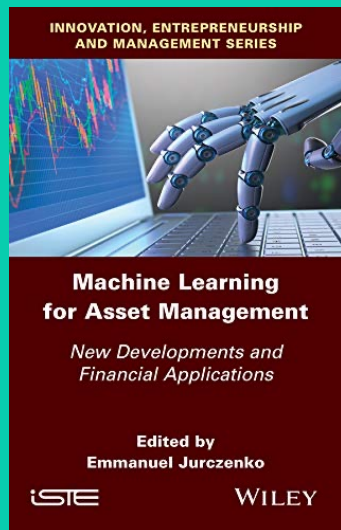
Determining the Risk-Free Rate

- The yield on U.S. Treasury securities
- Surveys suggest most practitioners use 10- to 30-year treasuries
- The Historical Risk Premium
- Estimate the risk premium ($E[R_{Mkt}] - r_f$) using the historical average excess return of the market over the risk-free interest rate

S&P 500 Excess Return Versus	Period	
	1926–2015	1965–2015
One-year Treasury	7.7%	5.0%
Ten-year Treasury*	5.9%	3.9%

12.3 Beta Estimation

Expected versus historical risk

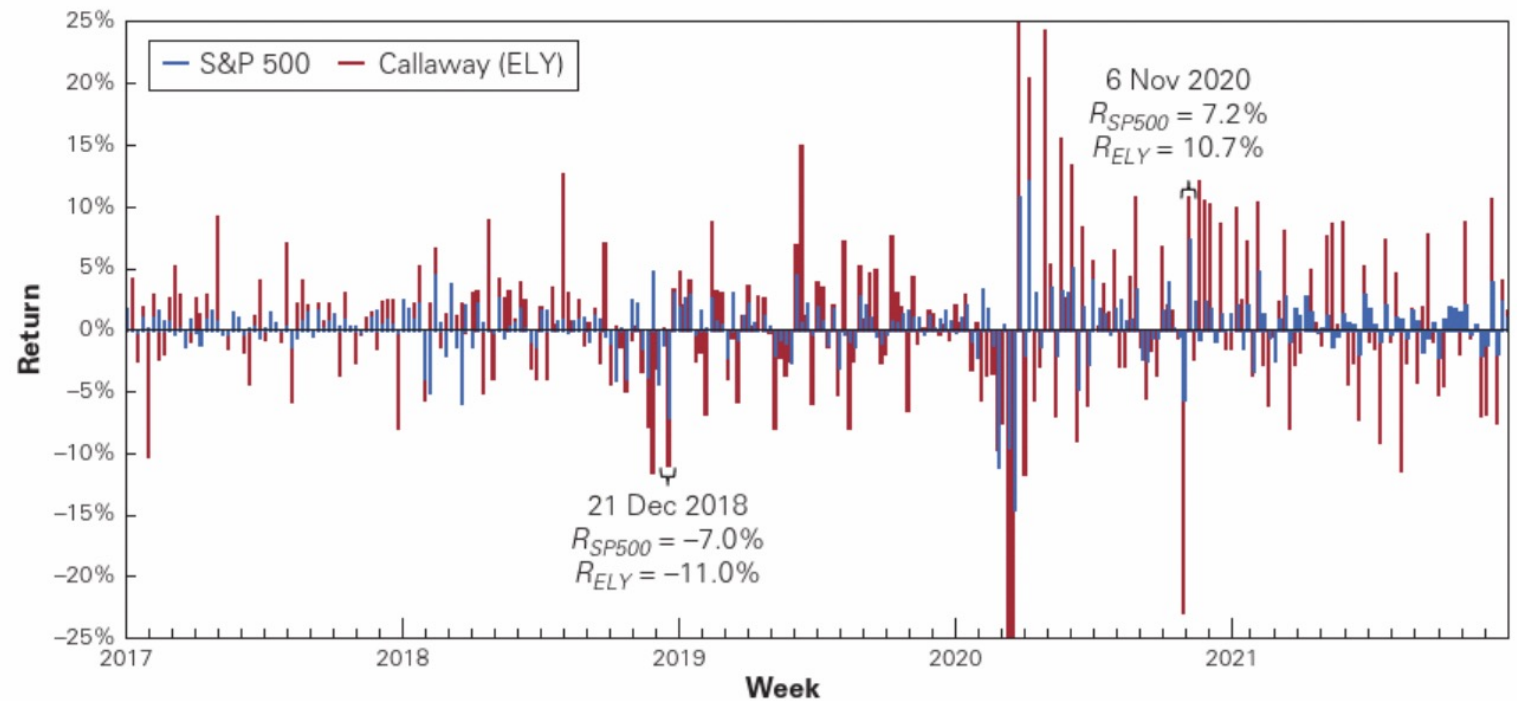


- Having identified a market proxy, the next step in implementing the CAPM is to determine the security's beta, which measures the sensitivity of the security's returns to those of the market.
- Because beta captures the market risk of a security, as opposed to its diversifiable risk, it is the appropriate measure of risk for a well-diversified investor.
- Ideally, we would like to know a stock's beta in **the future** ; that is, how sensitive will its future returns be to market risk.
- In practice, we estimate beta based on the stock's historical sensitivity. This approach makes sense if a stock's beta remains relatively stable over time, which appears to be the case for most firms.
- Optional Reading:
 - If you are interested in advances in beta forecasting, please see Chapter 7 "Forecasting Beta Using Machine Learning and Equity Sentiment Variables" by JOUROVSKI, DUBIKOVSKYY, ADELL, RAMAKRISHNAN and KOSOWSKI in the Wiley book 'Machine Learning for Asset Management'

Figure 12.1

Weekly Returns for Callaway
Golf's Stock and for the S&P
500, 2017–2021

Callaway's returns tend to move in the same direction, but with greater amplitude, than those of the S&P 500.



Now regress Callaway excess returns on the market excess returns.

Using Linear Regression

- Linear Regression

- The statistical technique that identifies the best-fitting line through a set of points.

$$(R_i - r_f) = \alpha_i + \beta_i(R_{Mkt} - r_f) + \varepsilon_i$$

- α_i is the intercept term of the regression
- $\beta_i(R_{Mkt} - r_f)$ represents the sensitivity of the stock to market risk. When the market's return increases by 1%, the security's return increases by $\beta_i\%$
- ε_i is the error term and represents the deviation from the best-fitting line and is zero on average

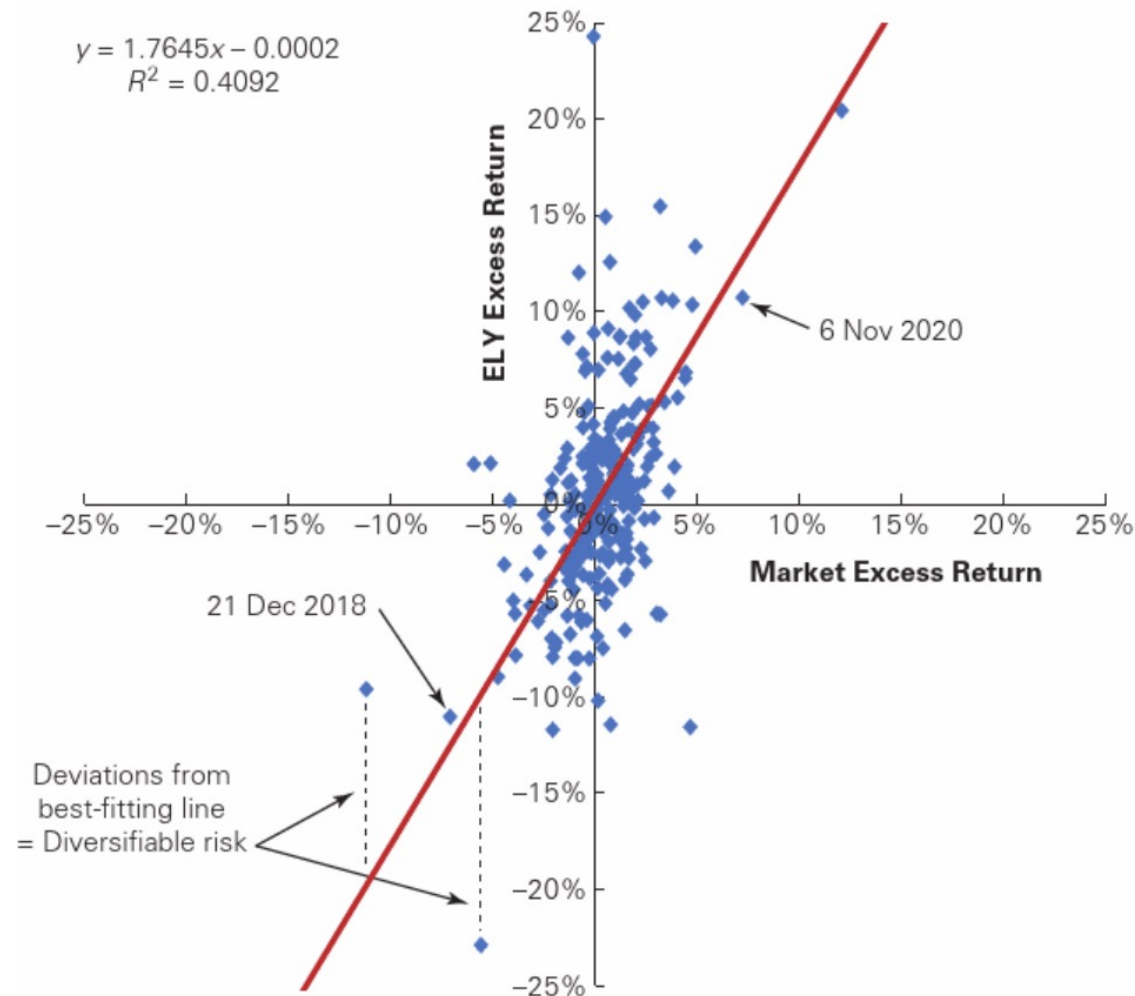
- Since $E[\varepsilon_i] = 0$:

$$E[R_i] = \underbrace{r_f + \beta_i(E[R_{Mkt}] - r_f)}_{\text{Expected return for } i \text{ from the SML}} + \underbrace{\alpha_i}_{\text{Distance above/below the SML}}$$

- α_i represents a risk-adjusted performance measure for the historical returns. If α_i is positive, the stock has performed better than predicted by the CAPM. If α_i is negative, the stock's historical return is below the SML.

Figure 12.2 Linear Regression and Scatterplot of Weekly Excess Returns for Callaway Golf Versus the S&P 500, 2017–2021

Beta corresponds to the slope of the best-fitting line. Beta measures the expected change in Callaway's excess return per 1% change in the market's excess return. Deviations from the best-fitting line correspond to diversifiable, non-marketrelated risk. In this case, Callaway's estimated beta is approximately 1.76.

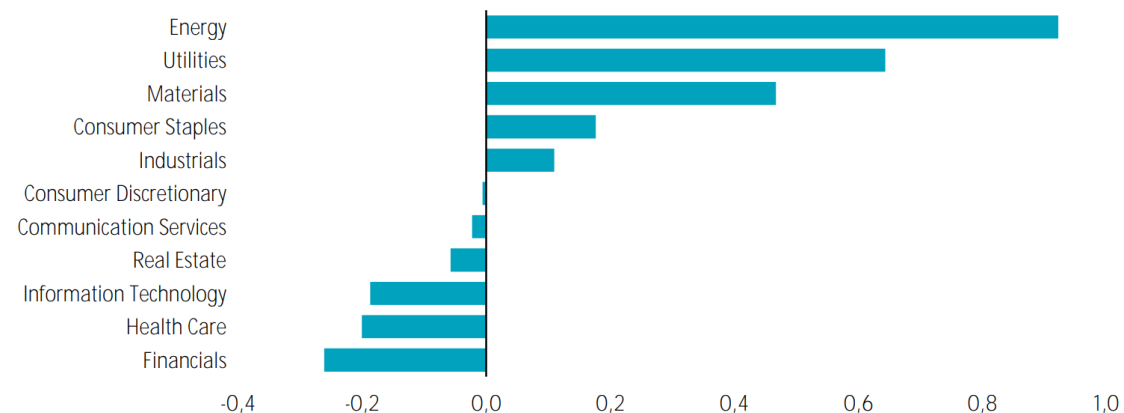


Application of Beta concept to Carbon/climate finance:

Robeco's Carbon Beta

- Robeco introduce a measure of asset-level climate risk exposure that is determined by how an asset's return is correlated with a carbon risk factor:
 - An asset with a negative carbon beta, for example, tends to have high returns when carbon risks increase.
 - Such an asset can be seen as a 'climate hedge' as it pays off in times when investors are especially concerned about climate change

Figure 1 | Average carbon beta per industry group



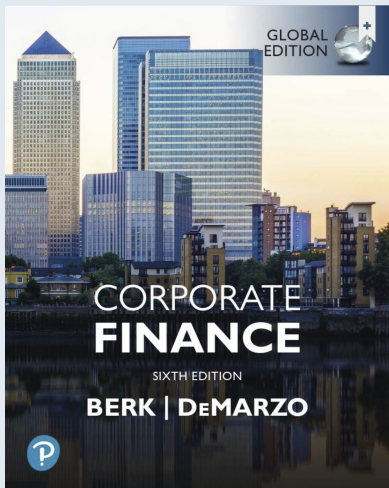
Source: The sample consists of common equities in the CRSP-Compustat universe traded on the NYSE, Nasdaq, or AMEX. The sample period is from January 2007 to December 2020.

Source: <https://www.robeco.com/files/docm/docu-202204-carbon-beta-is-an-additional-climate-risk-indicator-above-and-beyond-traditional-carbon-data.pdf>

Chapter 12 Quiz

1. What inputs do we need to estimate a firm's equity cost of capital using the CAPM?
2. How do you determine the weight of a stock in the market portfolio?
3. How can you estimate the market risk premium?
4. How can you estimate a stock's beta from historical returns?

Chapter 13 (BD)



Chapter Outline

- 13.1** Competition and Capital Markets
- 13.2** Information and Rational Expectations
- 13.6** Style-Based Techniques and the Market Efficiency Debate
- 13.7** Multifactor Models of Risk
- 13.8** Methods Used in Practice

Main Takeaways Week 4

- **BD Chapter 10**
 - Common measures of risk and return
 - Understanding Diversification, systematic return, beta and the cost of capital
- **BD Chapter 11**
 - How to calculate portfolio expected return, volatility and beta
 - Differentiating Capital Market Line from Security Market Line
 - Deriving the Capital Asset Pricing Model
- **BD Chapter 12**
 - How to Estimate the Market Portfolio and Beta
- **BD Chapter 12**
 - How to calculate alpha and understand the effect of competition in capital markets
 - Understanding how to test the CAPM
 - Style-Based Techniques and Multi-factor models of risk

Any questions?

Appendix:

Expected Returns and Beta

Problem

Suppose the risk-free rate is 5% and the economy is equally likely to be strong or weak.

Use Eq. 10.11 to determine the cost of capital for the type S firms considered in Example 10.8.

How does this cost of capital compare with the expected return for these firms?

Textbook Example 10.9

Solution

Solution

If the economy is equally likely to be strong or weak, the expected return of the market

and the market risk premium is

Given the beta of for type S firms that we calculated in Example 10.8, the estimate of the cost of capital for type S firms from Eq. 10.11 is

Thus, investors who hold these stocks can expect a return that appropriately compensates them for the systematic risk they are bearing by holding them (as we should expect in a competitive market).

Appendix: Diversification using Different Types of Stocks

Example 11.7

Problem

Stock within a single industry tend to have higher correlation than stocks in different industries.

Likewise, stocks in different countries have lower correlation on average than stocks within the United States.

What is the volatility of a very large portfolio of stocks within an industry in which the stocks have a volatility of 40% and a correlation of 60%?

What is the volatility of a very large portfolio of international stocks with a volatility of 40% and a correlation of 10%?

Diversification using Different Types of Stocks

Solution

Example 11.7

Appendix (optional
and not
examinable)

CAPM Tests

What the CAPM
Predicts And How
to Test It

Essence of CAPM:

- Expected return on any asset is a positive linear function of its beta and that beta is the only measure of risk needed to explain the cross-section of expected returns.

How to test it:

- The basic idea of the approach is
 - the use of **a time series** (first pass) regression to estimate betas and
 - the use of **a cross-sectional** (second pass) regression to test the hypothesis derived from the CAPM

Appendix:

CAPM Test Methodology: Two- Pass Regression

Step 1 (First-Pass Regression):

For each of the N securities included in the sample, we first run the following regression over time to estimate beta:

$$R_{it} = r_{it} - r_{ft} = \alpha_i + \beta_i R_{mt} + e_{it} \quad (1)$$

Step 2 (Second-Pass Regression)

We run the following cross-section regression over the sample period over the N securities:

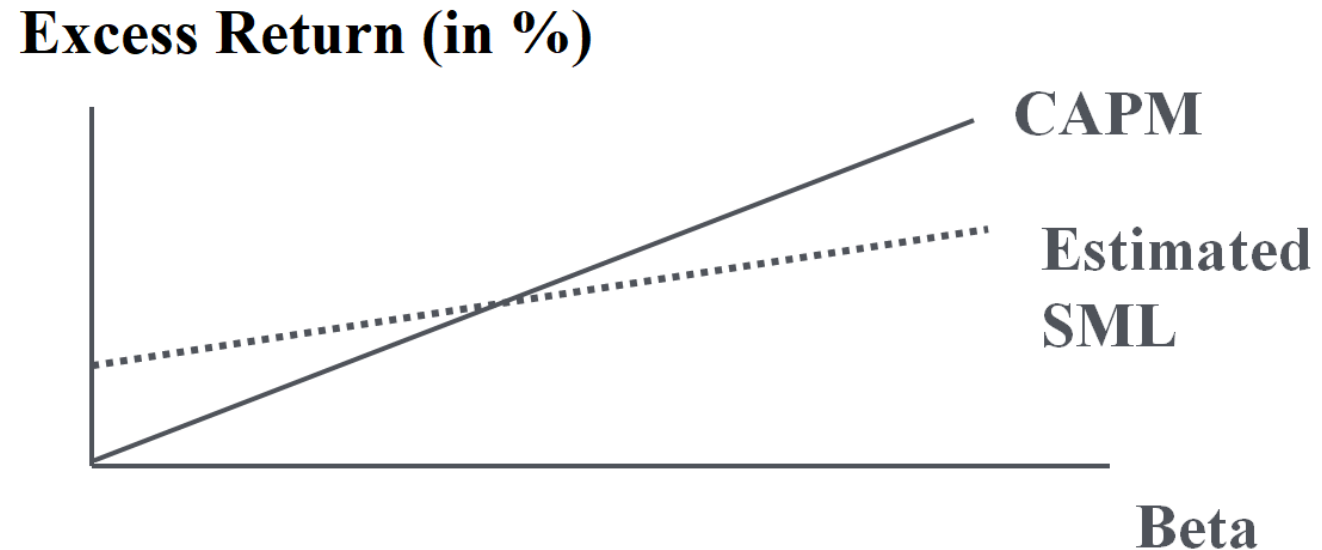
$$\bar{R}_i = \overline{r_i - r_f} = \gamma_0 + \gamma_1 \beta_i \quad \text{for } i=1, \dots, N (\text{say 100 stocks}) \quad (2)$$

$$\bar{R}_i = \overline{r_i - r_f} = \gamma_0 + \gamma_1 \beta_i + \gamma_2 \sigma^2(e_i) \quad (2 \text{ extended})$$

The coefficient γ_1 captures if there is a relationship between cross-sectional differences in returns and cross-sectional differences in market beta

Appendix:

Test Result: SML slope is “too flat” and intercept is “too high”.



- The above linear **cross-sectional** regression tests the relationship between expected returns and CAPM beta
- One extension is to test if other variables such as valuation ratios (e.g. B/M) explain cross-sectional differences in returns. We will return to when discussing multi-factor models in BD 13



Clifford Asness ✓
@CliffordAsness

I have a joke about the CAPM but the securities market punchline fell flat.

1:31 am · 27 Jul 2020

Appendix:

Extending the Hypotheses Tested and Adding Additional Regressors: Fama and MacBeth (1973) Study

- Fama and MacBeth (1973) estimate:
 - rank portfolios by betas in the second step to create large dispersion in betas
 - estimate SML for portfolios of stocks (Black, Jensen and Scholes (1972))

$$\overline{r_i - r_f} = \gamma_0 + \gamma_1 b_i + \gamma_2 b_i^2 + \gamma_3 \sigma(e_i) \quad (13.4)$$

Table 13.1 Risk and return in the Fama-MacBeth study. All rates are in basis points per month. Sample period was 1935–June 1968.

Period	1935/6–1968
Average γ_0	8
t -statistic (testing $\gamma_0 = 0$)	0.20
Average $r_M - r_f$	130
Average γ_1	114
t -statistic (testing $\gamma_1 = r_M - r_f$)	1.85
Average γ_2	-26
t -statistic (testing $\gamma_2 = 0$)	-0.86
Average γ_3	516
t -statistic (testing $\gamma_3 = 0$)	1.11
Average R -square	0.31

- Another extension is to examine whether in addition to CAPM beta other variables such as, for example, emissions or other emissions related disclosure explain cross-sectional differences in returns
- You will study these findings in the course 'Financial Markets and Climate Change'